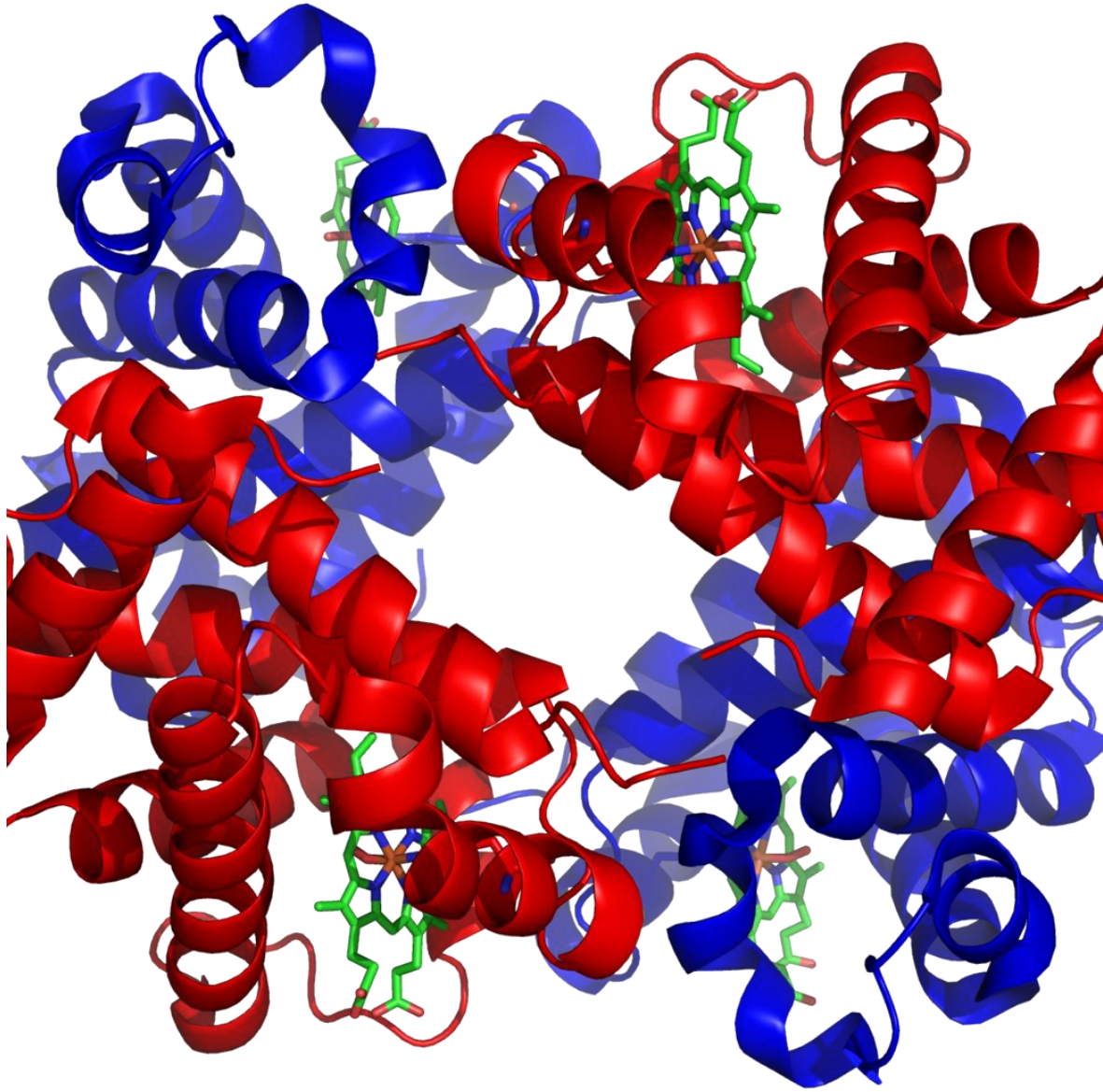
DP Biology – B1.2

Proteins

- How does the order of amino acids produce such a variety of protein forms and functions?
- How do chemical and physical conditions affect protein molecules?

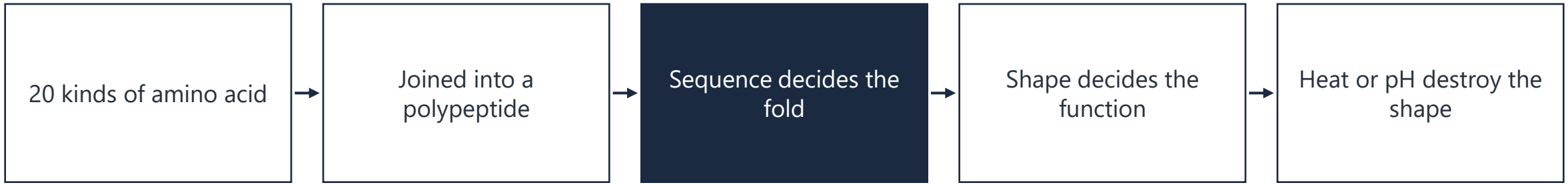


SL statements B1.2.1–5. Textbook pp. 205–210.
Image: model of haemoglobin. Zephyris, CC BY-SA 3.0.

Objectives for each class

Class	Statements	By the end of the class, you can...
1	B1.2.1–2	Draw a generalized amino acid, and show how condensation forms a dipeptide and then a polypeptide
2	B1.2.3–4	Explain dietary needs for amino acids, and why the variety of peptide chains is almost limitless
3	B1.2.4–5	Explain how sequence decides shape, and how heat and pH denature proteins

From sequence to function



Proteins speed up reactions, carry oxygen, fight infection and give strength. All are built from the same 20 **amino acids**.

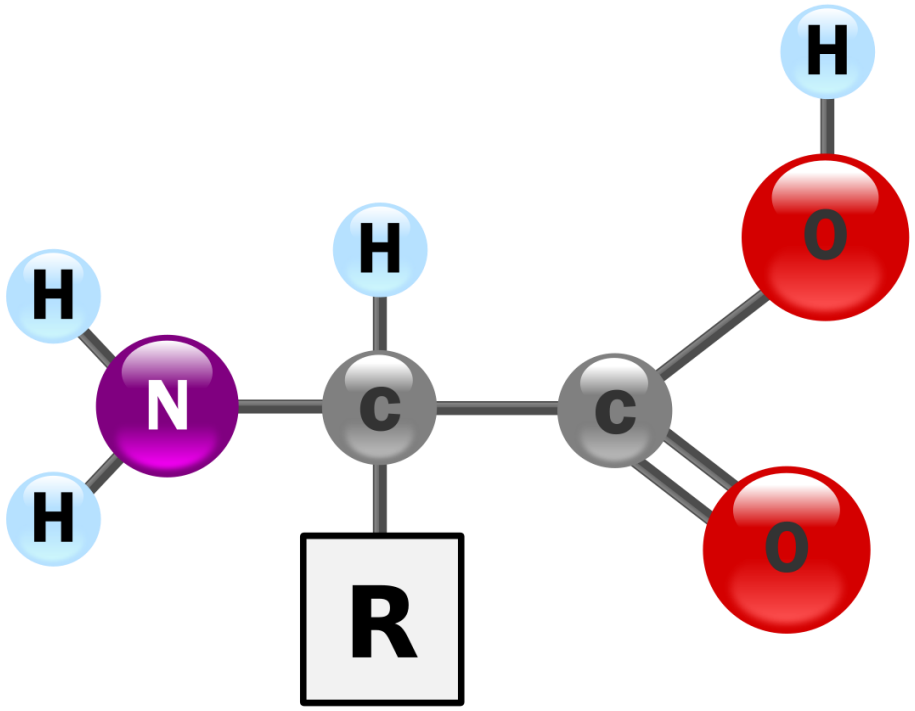
The **order** of amino acids makes each chain fold into one shape for one job.

Big question

How can the same 20 building blocks make an enzyme, an antibody, a hormone and a hair?

The structure of an amino acid

B1.2.1



General structure of an amino acid. YassineMrabet, public domain.

A central **alpha carbon** carries four groups: an **amine group** (-NH_2), a **carboxyl group** (-COOH), a **hydrogen** and an **R-group**.

The R-group is different in each of the 20 amino acids.

Exam tip

Draw it with R for the side chain, amine on the left and carboxyl on the right.

R-groups make each amino acid different

B1.2.1

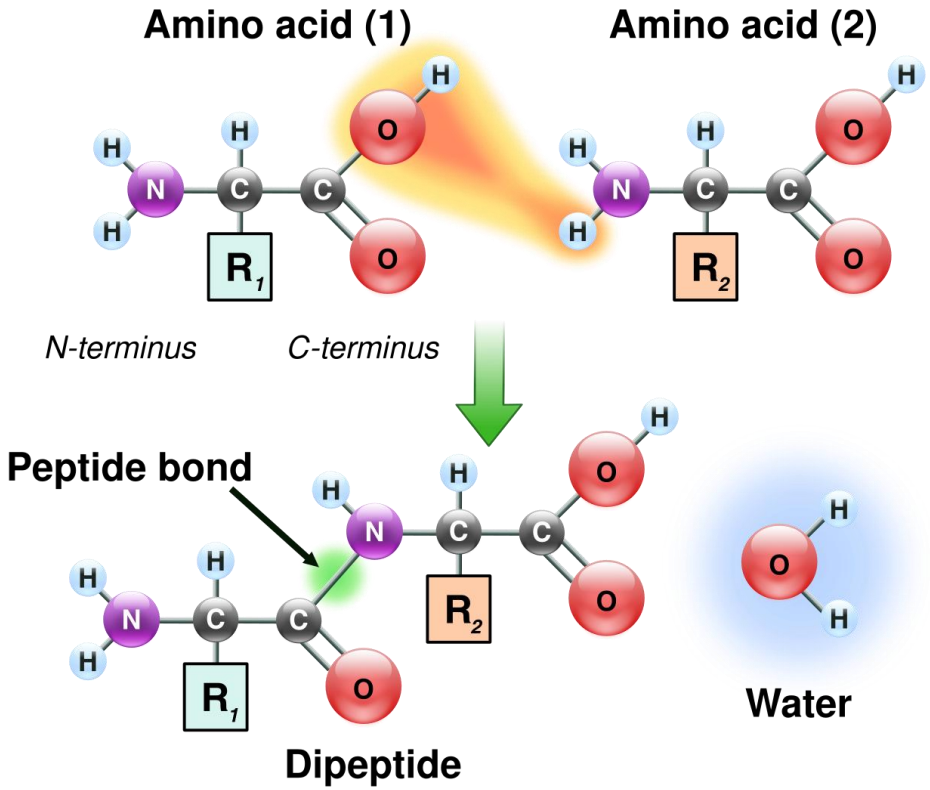
R-group	Example	In a protein
Non-polar	Alanine	Hydrophobic: buried inside
Polar	Serine	Hydrophilic: bonds with water
Charged	Lysine, aspartic acid	Ionic attractions; affected by pH
Smallest	Glycine (-H)	Lets the chain bend tightly

Key idea

R-groups differ in size, charge and polarity, so each sequence folds its own way.

Condensation joins two amino acids

B1.2.2



Two amino acids form a dipeptide. YassineMrabet, public domain.

The **carboxyl** of one amino acid reacts with the **amine** of the next; -OH and -H leave as **water**.

The new C-N bond is a **peptide bond**: a **dipeptide**.

Word equation

amino acid + amino acid → dipeptide + water

From dipeptides to polypeptides

B1.2.2



each bead = one amino acid; each link = a peptide bond

Each amino acid is added by another condensation, building a **polypeptide**. In cells this happens on **ribosomes**, following mRNA.

Polypeptide	Amino acids	Function
Insulin (two chains)	21 and 30	Lowers blood glucose
Titin	about 34 000	Elastic spring in muscle

Common mistake

A protein is one or more polypeptides folded into a working shape.

Practice: build a peptide

B1.2.1–2

Draw

Draw a dipeptide; label the peptide bond and the water released.

Count

How many waters are released when 50 amino acids join?

Model

Use a model kit to join two amino acids.

Check

Peptide bond between C and N; water on the products side; amine end and carboxyl end labelled.

Check-in: class 1

B1.2.1–2

1. Name the four groups on the alpha carbon.
2. Draw a generalized amino acid.
3. Name the bond between two amino acids.
4. Outline how a dipeptide forms.
5. Distinguish between a polypeptide and a protein.

How to use this

Answer without your notes, then check against slides 3–8.

Essential and non-essential amino acids

B1.2.3

Food	Protein / %
Lentils (dry)	about 25
Chicken	about 27
Rice (dry)	about 7
Egg	about 13

Non-essential amino acids can be made in the body. **Essential** ones cannot and must come from food: nine for adults.

If one essential amino acid runs short, protein synthesis slows, however plentiful the rest.

Think

Why does a shortage of just one amino acid limit protein synthesis?

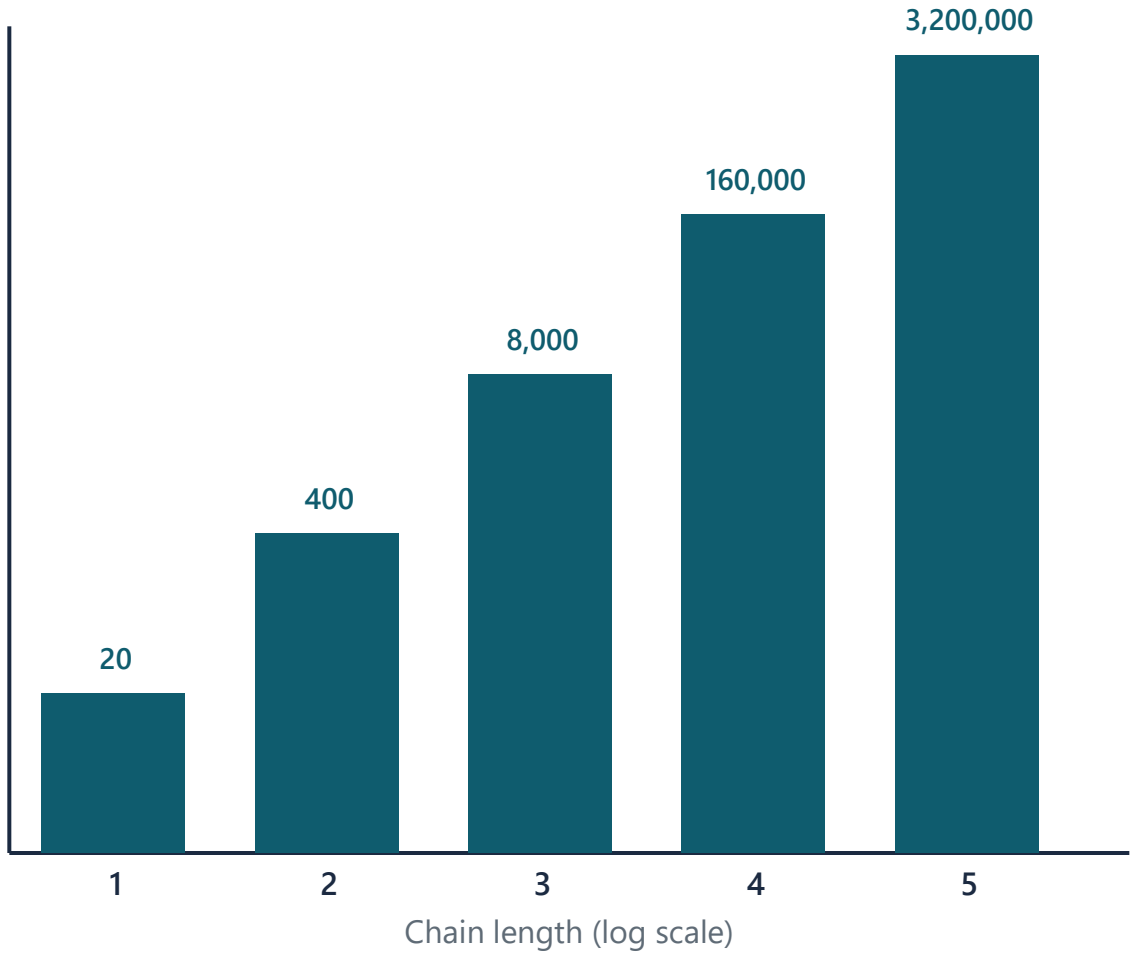
Plant-based diets

B1.2.3

Food group	Often low in	Good partner
Cereals	Lysine	Beans or lentils
Pulses	Methionine	Cereals, nuts, seeds
Soya, quinoa	Contain all essentials	

Vegans get all the essential amino acids by eating a **variety** of plant foods, such as rice with beans.

An almost limitless variety



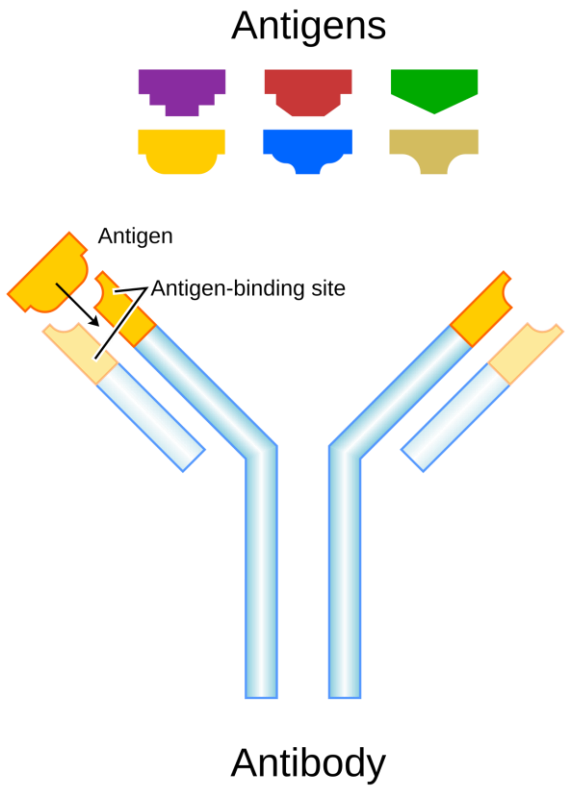
With 20 amino acids and any order, a chain of **n** has **20 to the power n** possible sequences.

Chains run to thousands of amino acids, so the variety is limitless in practice.

Examples of proteins and their functions

B1.2.4

Protein	Function
Haemoglobin	Carries oxygen
Antibodies	Bind specific antigens
Rhodopsin	Absorbs light in the retina
Collagen	Strength in skin, tendons, bone
Insulin	Hormone: cells take in glucose



An antibody. Fvasconcellos, public domain.

Big question: same 20 building blocks, different sequences, different shapes, different jobs.

Practice: diets and sequences

B1.2.3–4

Count

How many sequences for a peptide of 3 amino acids?

Diet

A diet is only rice. Which essential amino acid may run short?

Compare

How does this compare with DNA's 4 to the power n?

Explain

Give a reason for each answer.

Check-in: class 2

B1.2.3–4

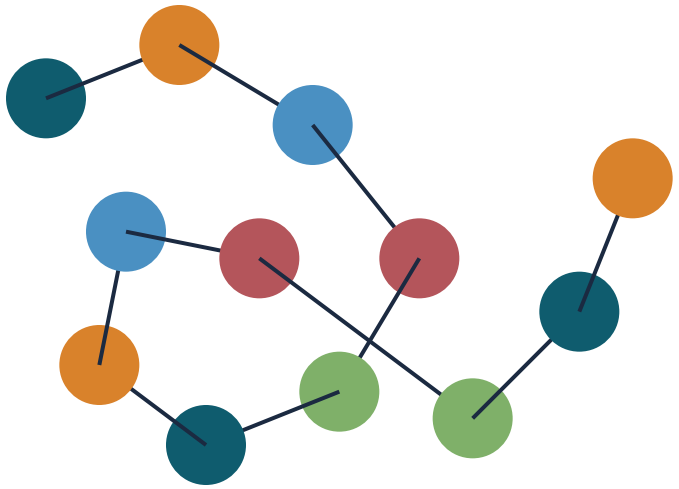
1. Distinguish essential from non-essential amino acids.
2. Explain why vegans should eat a variety of plant proteins.
3. Calculate the sequences possible for 4 amino acids.
4. Give three proteins and their functions.
5. Explain why the variety of proteins is so great.

How to use this

Answer without your notes, then check against slides 10–14.

Sequence determines shape

B1.2.4



the chain folds as R-groups interact

Non-polar R-groups cluster in the middle, away from water; charged and polar ones attract each other.

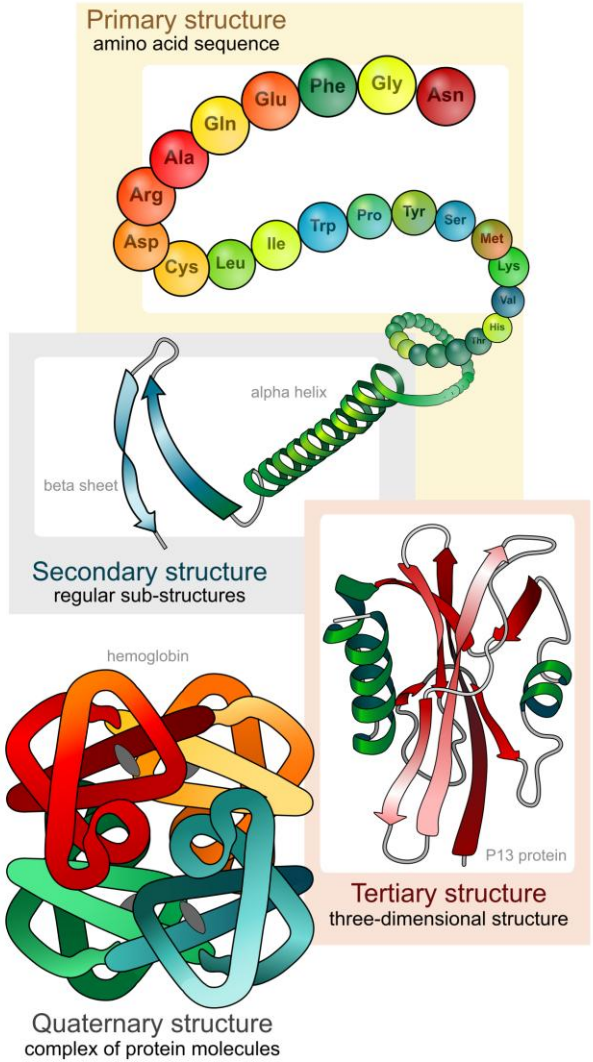
So a sequence **always folds the same way**. Change one amino acid and the shape, and function, may change.

Link

HL: levels of structure. D1.2: a single change in sickle cell haemoglobin.

Shapes of real proteins

B1.2.4



Levels of protein structure. LadyofHats, public domain.

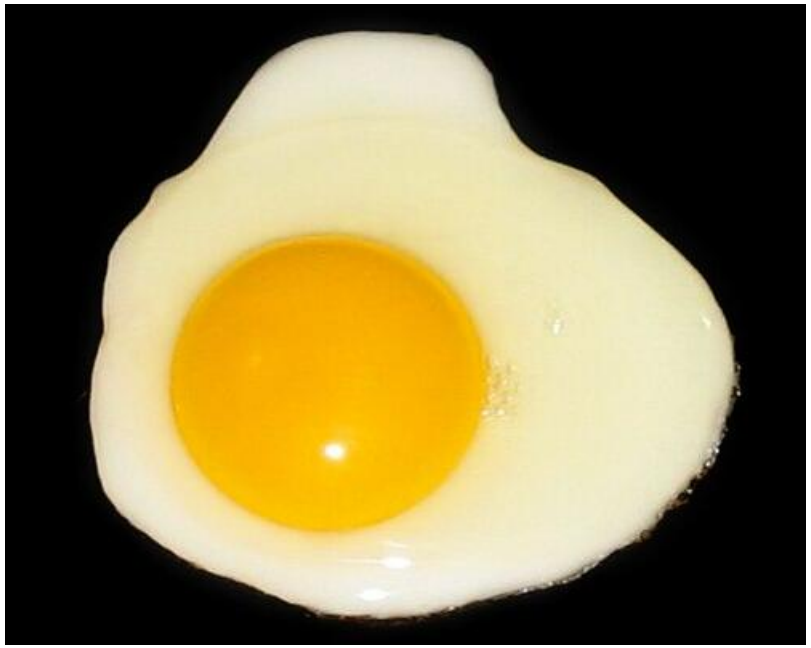
Folded chains form coils and sheets, which fold further into a 3D shape. Some proteins join several chains together (haemoglobin has four).

HL

Primary, secondary, tertiary and quaternary structure are named at HL (B1.2.7–11).

Denaturation by heat

B1.2.5



Cooked egg: heat has denatured the albumin. David Benbennick, public domain.

Heat makes the protein vibrate until the weak bonds holding its shape break and it **unfolds**: it is **denatured** and cannot work. Usually it cannot refold.

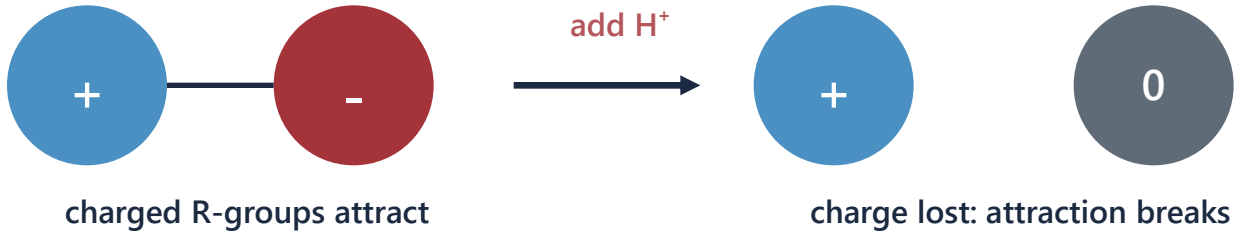
Egg white turns white and solid as its proteins denature and tangle.

Common mistake

Denaturation does not break peptide bonds: the sequence stays the same; only the shape is lost.

Denaturation by pH

B1.2.5



Changing the pH adds or removes H^+ ions, so charged R-groups change. The attractions holding the shape break and the protein unfolds.

Protein	Best pH	Shows
Pepsin (stomach)	about 2	Adapted to strong acid
Trypsin (small intestine)	about 8	Adapted to alkaline conditions

Key idea

Each protein keeps its shape only in a range of pH and temperature.

Working with data: heating egg white

B1.2.5

Egg white held 10 minutes at each temperature; cloudiness measured with a colorimeter (higher absorbance = more denatured). Invented values; realistic pattern.

Temperature / °C	Absorbance
40	0.02
50	0.03
60	0.21
70	0.64
80	0.95
90	0.97

1. Describe the effect of temperature.
2. Explain the change from 50 to 80 °C.
3. Suggest why it levels off above 80 °C.
4. State two control variables.

Practice: explain the observation

B1.2.5

Fever

Why is a body temperature of 42 °C dangerous?

Stomach

Why does salivary amylase stop working in the stomach?

Hot springs

How can some bacteria live at 80 °C?

Explain

Use the words shape, denature and R-groups.

Check-in: class 3

B1.2.4–5

1. Explain how the sequence of amino acids decides a protein's shape.
2. Define denaturation.
3. Explain why a boiled egg cannot become raw again.
4. Explain how pH can denature a protein.
5. State what is not changed by denaturation.

How to use this

Answer without your notes, then check against slides 16–21.

Common mistakes to avoid

Not this	But this
A peptide bond joins two R-groups.	It joins the carboxyl C of one to the amine N of the next.
Hydrolysis forms a dipeptide.	Condensation forms it; hydrolysis breaks it.
Polypeptide and protein mean the same.	A protein is folded polypeptide(s).
Denaturation breaks proteins into amino acids.	It changes shape; the sequence stays.
Essential amino acids are the most important.	Essential means they must come from food.

Summary: proteins

B1.2.1–5

	Key idea
B1.2.1	Amino acid: amine, carboxyl, H, R-group
B1.2.2	Condensation forms peptide bonds: polypeptides
B1.2.3	Essential amino acids from diet; varied plant foods
B1.2.4	20 to the power n sequences; sequence decides shape
B1.2.5	Heat and pH denature proteins

Check that you can...

1. **Draw** a generalized amino acid and a dipeptide
2. **Distinguish** between essential and non-essential amino acids
3. **Calculate** possible peptide sequences
4. **Explain** how sequence determines shape
5. **Explain** how heat and pH denature proteins

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Name the four groups attached to the alpha carbon of an amino acid.
2. Calculate the number of possible sequences for a peptide of 3 amino acids.
3. Explain why a boiled egg cannot go back to being raw.

Image credits

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Peptidformationball.svg – GYassineMrabet – Public domain
Antibody.svg – Fvasconcellos 19:03, 6 May 2007 (UTC) – Public domain
Main protein structure levels en.svg – LadyofHats – Public domain
Fried egg, sunny side up.jpg – David Benbennick – Public domain

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